教學實作：擴充 DHT11 溫濕度感測器功能

步驟 1：修改硬體定義提示詞 (Devices Definition)

讓 AI 認識 DHT11 感測器及其物理特性。

AMB82-mini ： PIN 8
HUB 8735 Ultra ：PIN 20

DHT11 Temperature & Humidity Sensor
- Pin mapping: depends on development board
    - AMB82-mini: PIN 8
    - HUB 8735 Ultra: PIN 20
- Measures: temperature (°C) and relative humidity (%)
- Read mode: single trigger, returns two integer values
- Temperature range: 0–50 °C
- Humidity range: 20–90 % RH
- Physical Rules: Values are integers. Sensor requires ~1 s between reads.

步驟 2：定義工具 JSON 規範 (Tools Definition)

在 toolsDefinition 中加入 /dht11 指令的格式規範。

Reading the DHT11 temperature and humidity sensor:

```
{
  "type": "tool_call",
  "method": "/dht11",
  "params": {
    "pin": "<Device pin number. If the user does not specify a pin, ask first.>"
  }
}
```

Success response:

```
{
  "status": "success",
  "method": "/dht11",
  "temperature": <temperature value>,
  "humidity": <humidity value>,
  "workId": "<system-provided>"
}
```

Error response:

```
{
  "status": "error",
  "method": "/dht11",
  "reason":"<error reason>",
  "workId": "<system-provided>"
}
```

步驟 3：底層 C++ 程式碼實作

在專案中新增 tool_dht11 驅動函數。

```
#include "DHT.h"
#define DHTPIN 20
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

String tool_dht11(int pin, String workId) {
  float h = dht.readHumidity();
  float t = dht.readTemperature();
  if (isnan(h) || isnan(t)) {
    return "{\"status\":\"error\","
           "\"method\":\"/dht11\","
           "\"reason\":\"dht11_read_failed\","
           "\"workId\":\"" + workId + "\"}";

  }

  return
    "{\"status\":\"success\","
    "\"method\":\"/dht11\","
    "\"temperature\":" + String(t) + ","
    "\"humidity\":"    + String(h) + ","
    "\"workId\":\"" + workId + "\"}";
}

void setup() {
  // ... 原有初始化內容 ...
  delay(1000);  // 等待感測器穩定

  dht.begin();

}
```

步驟 4：更新工具分發器邏輯 (executeTool)

將 /dht11 方法註冊進系統的分發流程。

```
void executeTool(String workId, String command, JsonObject params, bool reCheck = true) {

  if (command == "/digitalwrite" || command == "/analogwrite") {
    // ... 原有內容不變 ...

  }

  else if (command == "/dht11") {
      int pin = params["pin"].as<int>();

      String response = tool_dht11(pin);

      historicalMessages += buildGeminiMessage("user", command + timestamps);
      historicalMessages += buildGeminiMessage("model", response + timestamps);

      executeToolHistory += workId + " " + command + " [ " + String(pin) + " | " + response    + " ]\n";

      evaluateWorkflowContinuation(workId, reCheck);
  }

  else if (command == "/digitalread" || command == "/analogread") {
    // ... 原有內容不變 ...
  }
  // ... 其餘 tools 不變 ...
}
```